

# Agentic Quant Lab

From AI-generated hypotheses to  
evidence-backed trading strategies

**Agents research.**  
**Deterministic systems decide.**



VISION → DISCIPLINED RESEARCH

# AI can generate trades. Can we trust them?

## TYPICAL AI TRADING BOT

LLM

“Bullish”

BUY

## AGENTIC QUANT LAB

\*Future only. Today ends at dry run.

Hypothesis

Evidence

Falsification

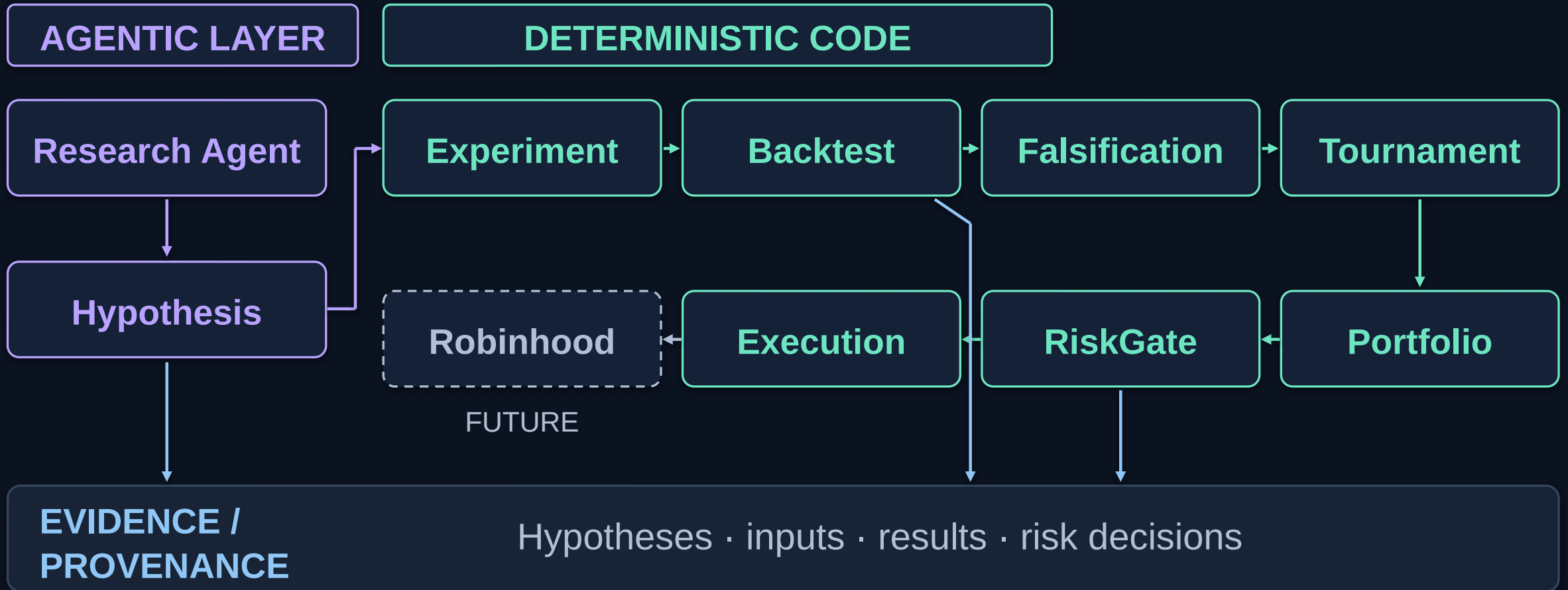
RiskGate

Trade\*

Hallucinated reasoning   Backtest overfitting   Look-ahead / bad data   Uncontrolled execution

The hard problem isn't generating strategies.  
It's rejecting bad ones.

# An autonomous research loop



Target architecture. Scripted agent today; demo receipts ≠ Phase 0 ledger.

# The agent has to prove itself wrong

## NULL CONTROL

0

No edge?  
No alpha.

## PLANTED SIGNAL

+

Known edge?  
Recover it.

## LEAKAGE TRAP

t+1

Future data?  
Reject it.

Costs

Frozen specifications

Regime tests

Data provenance

**Reject a failing rule. Don't optimize a rescue.**

Research requirements, not a claim that the demo validates market alpha.

# The system already says “no”

|                       |                |                                      |
|-----------------------|----------------|--------------------------------------|
| ETF Trend             | RESEARCH MORE  | Weak vs static de-risking            |
| Stock Momentum        | DATA BLOCKED   | Universe / identity / delistings     |
| Momentum + Quality    | DATA BLOCKED   | Stock history + PIT fundamentals     |
| PEAD / Earnings       | DEFER          | Consensus vintages unavailable       |
| ETF Relative Momentum | NOT YET TESTED | Frozen protocol; acquisition blocked |

**THIS IS THE POINT.**      The lab preserves negative evidence.

Trend: near rejection. Missing data is not an economic rejection.

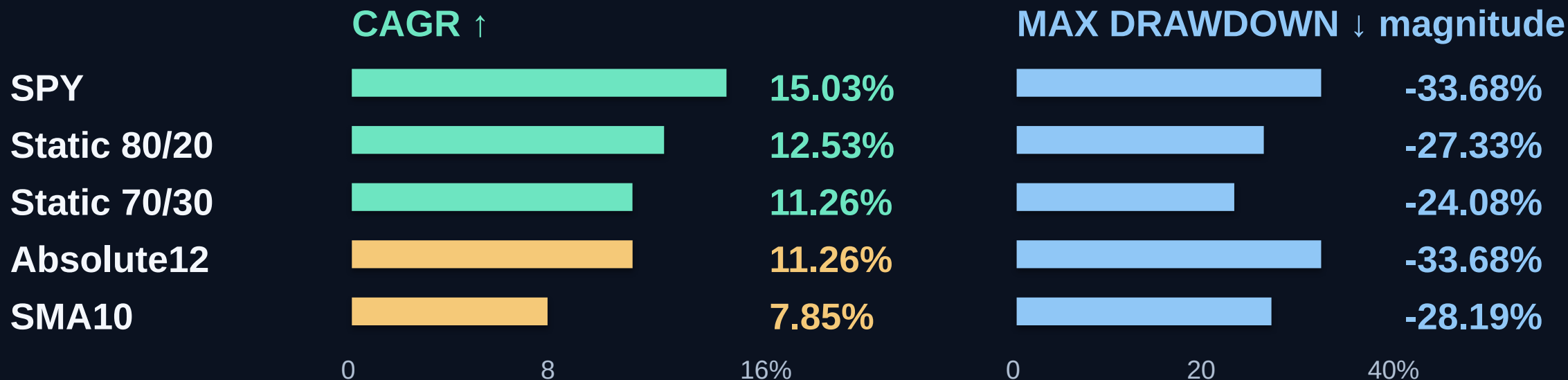
We didn't retune the failed rule.

# Trend looked promising — until we tested it properly

REAL RESEARCH · EXPLORATORY / NAV-BASED

NOT PRODUCTION-READY

2016-01-04 → 2026-07-31 · SPY / BIL · 20 bps one-way costs



Absolute12: no worst-drawdown benefit

SMA10: COVID protection, then missed rebound

Losing defensive cycles: Absolute12 4/4 · SMA10 9/11

# One command → research to dry-run execution

```
$ uv run aql demo
```

Agent → Experiment → Backtest

Falsification → Tournament → Portfolio

RiskGate → DRY RUN

**DATA:**  
**DEMO FIXTURE**

**LIVE TRADING:**  
**OFF**

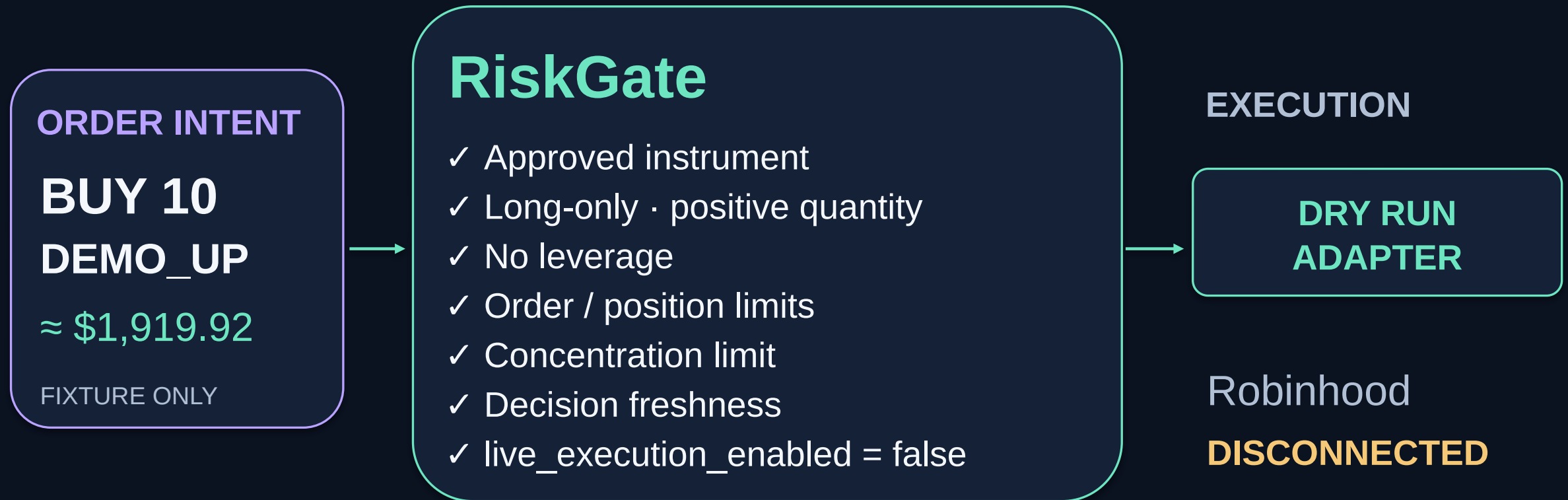
**EXECUTION:**  
**DRY RUN**

**INTERNET:**  
**NOT REQUIRED**

Scripted agent. Synthetic results. No LLM or broker call.

Prepared offline: `uv run --offline --no-sync aql demo`

# Agents never get direct control of capital



**Agents propose. RiskGate authorizes.**

One BUY / supplied demo state. No broker reconciliation.

Future design: credentials stay outside research agents.

# Where AI actually adds value

## AGENT / TARGET CAPABILITIES

- Read research
- Generate hypotheses
- Find failure regimes
- Interpret earnings / guidance
- Propose experiments
- Investigate degradation

## DETERMINISTIC CODE

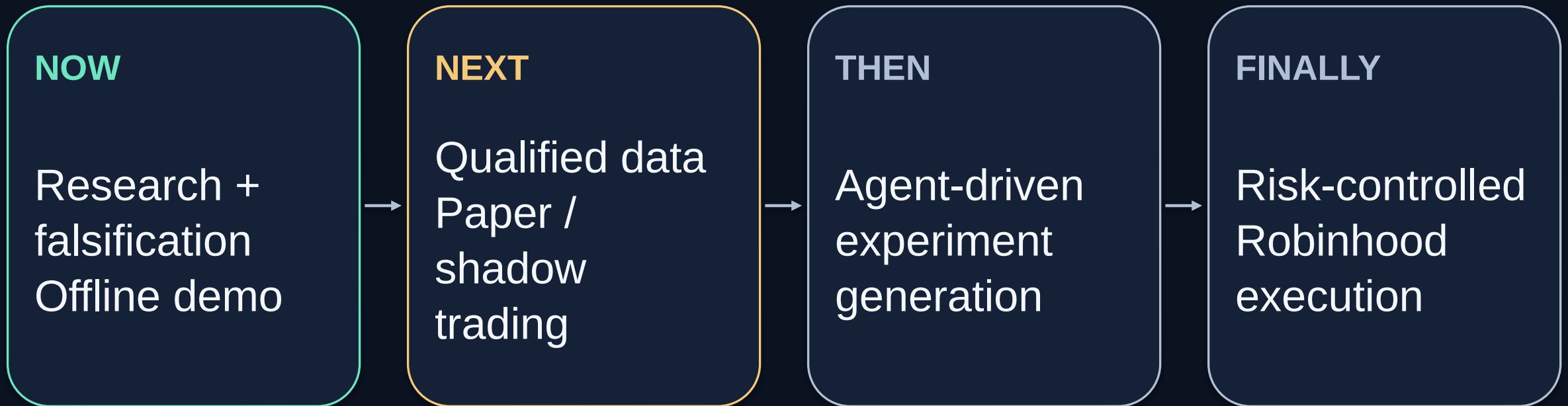
- Calculate returns
- Rank assets
- Run backtests + model costs
- Allocate capital
- Enforce risk limits
- Submit orders (future only)

**Use AI for reasoning.**

**Use code for truth.**

“Truth” means reproducible calculations, not certainty about markets.

# Roadmap



## LIVE TRADING DISABLED TODAY

Promotion requires evidence, not a better story.

The goal isn't an AI that trades more.

**It's an AI that knows  
when NOT to trade.**

RESEARCH

EVIDENCE

FALSIFICATION

RISK

**Agentic Quant Lab**

[github.com/BruceGK/agentic-quant-lab](https://github.com/BruceGK/agentic-quant-lab)